$F_p = 1048\,MHz$

$Z_L = 126{,}2 - j\,50{,}1$

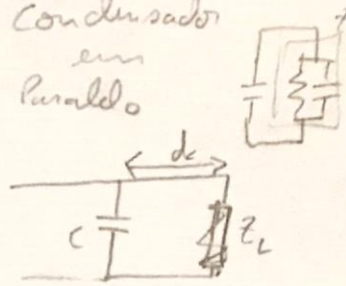
$j\ell = 2{,}504 - j\,1{,}002$

$Y_\ell = 0{,}34 + j\,0{,}14$ (C.S)

$Y_{\pi} = 1 + j\,1{,}15$

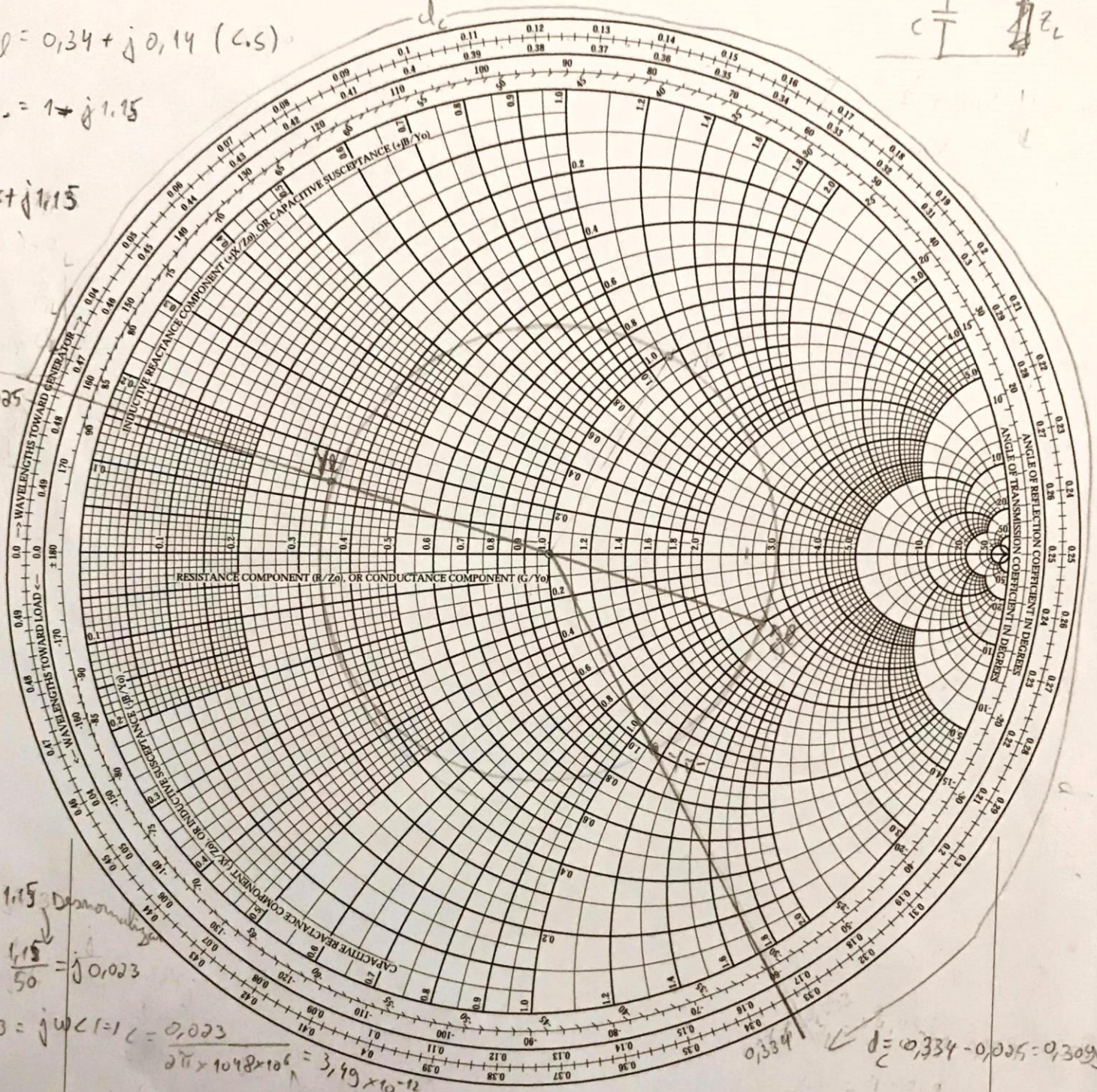
$Y_C = +j\,1{,}15$

Condensador em Paralelo



The Complete Smith Chart

Black Magic Design



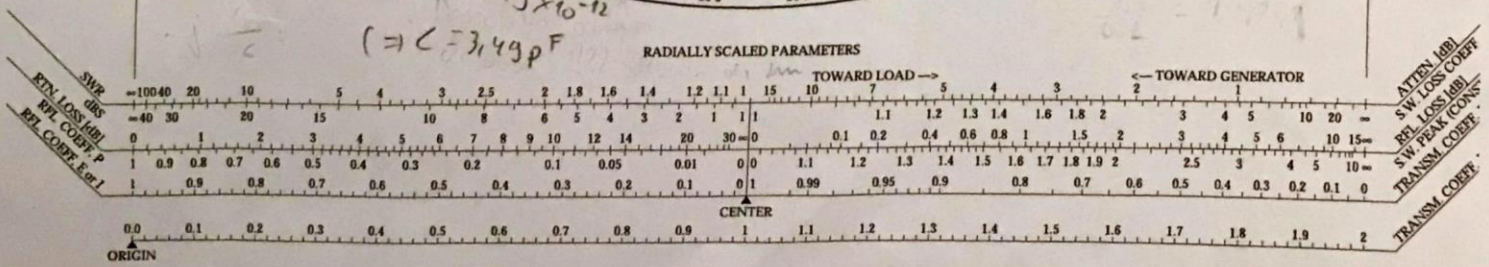
$Y_C = j\,1{,}15$ Desnormalizar

$Y_C = j\,\frac{1{,}15}{50} = j\,0{,}023$

$j0{,}023 = j\omega C \Rightarrow C = \frac{0{,}023}{2\pi \times 1048 \times 10^6} = 3{,}49 \times 10^{-12}$

$(\Rightarrow C = 3{,}49\,pF$

$d_c = 0{,}334 - 0{,}025 = 0{,}309\lambda$



$Y_\ell = 0{,}34 + j\,0{,}14$

Saber R_L:

$R_\ell = \frac{1}{0{,}34} = 2{,}941\,\Omega$

Desnormalizar

$R_L = 2{,}941 \times 50 \approx 147{,}06\,\Omega$ / No Netlab $R_L = 145{,}248\,\Omega$

Saber C_L:

$j\,0{,}14 = j\omega C_\ell$

$(\Rightarrow C_\ell = 2{,}126 \times 10^{-11}\,F$

$\Rightarrow C_L = 4{,}252 \times 10^{-13}\,F$

$(\Rightarrow C_L = 0{,}425\,pF$

No Netlab $C_L = 0{,}418\,pF$

$F_p = 1048\ MHz$

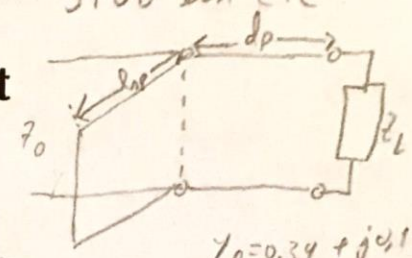
$Z_L = 125,2 - j\,50,1$

$y_l = 2,504 - j\,1,002$

$d_p = 0,166 - 0,024 = 0,142\,\lambda$

$l_{sp} = 0,364 - 0,25 = 0,114\,\lambda$

STUB en C.C



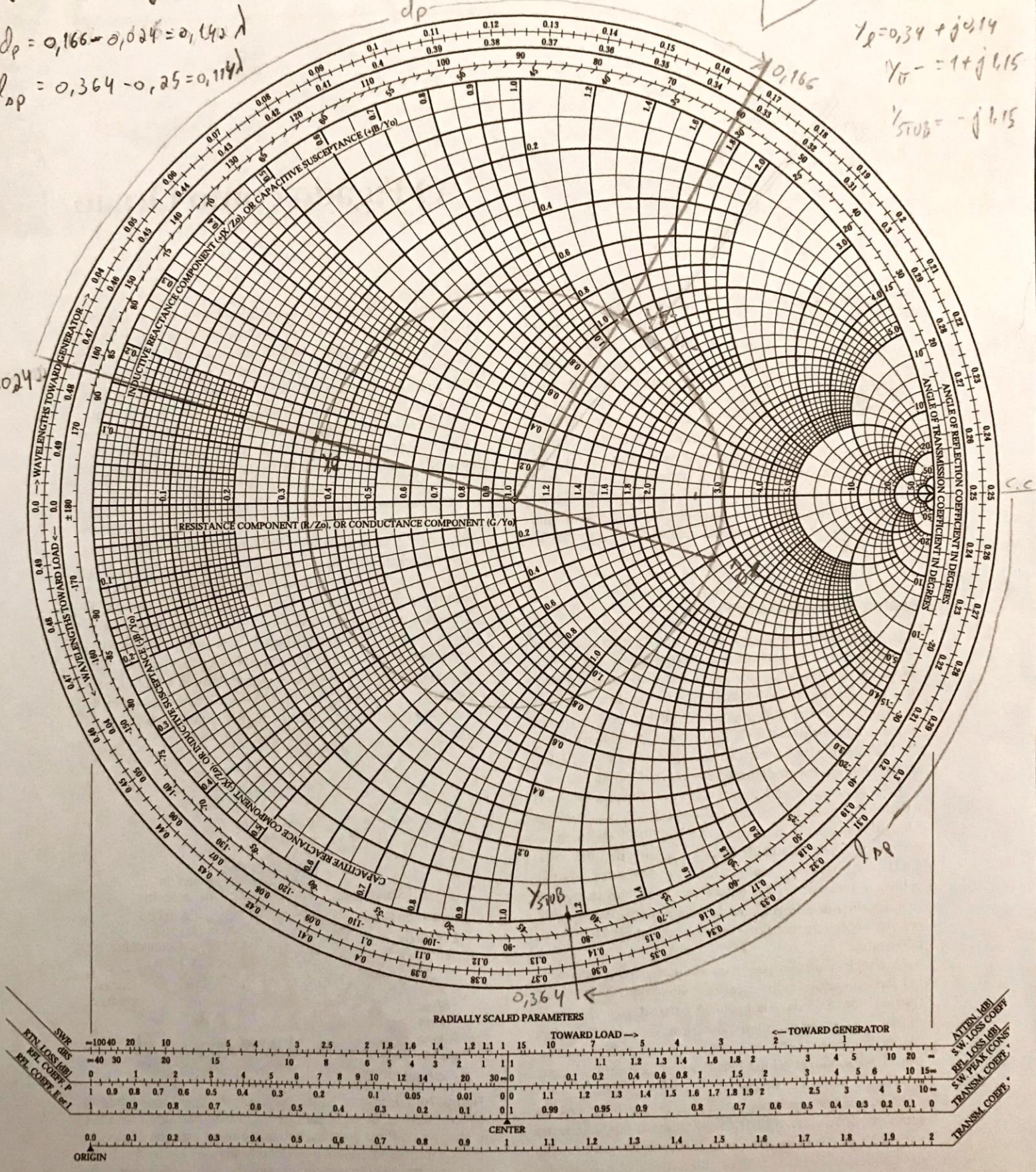
$Y_l = 0,34 + j\,0,14$

$Y_\pi = 1 + j\,1,15$

$Y_{STUB} = -j\,1,15$

The Complete Smith Chart

Black Magic Design



$F_p = 1048\ MHz$

$Z_L = 125{,}2 - j\,50{,}1$

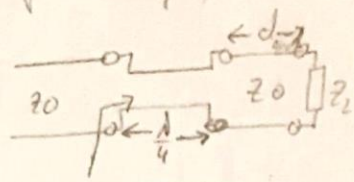
$z_p = 2{,}504 - j\,1{,}002$

$z_{mínimo} = 0{,}34\,\Omega$

desnormalizar

$z_{min} = 0{,}34 \times 50 = 17\,\Omega$

Transformador $\frac{\lambda}{4}$, num mínimo

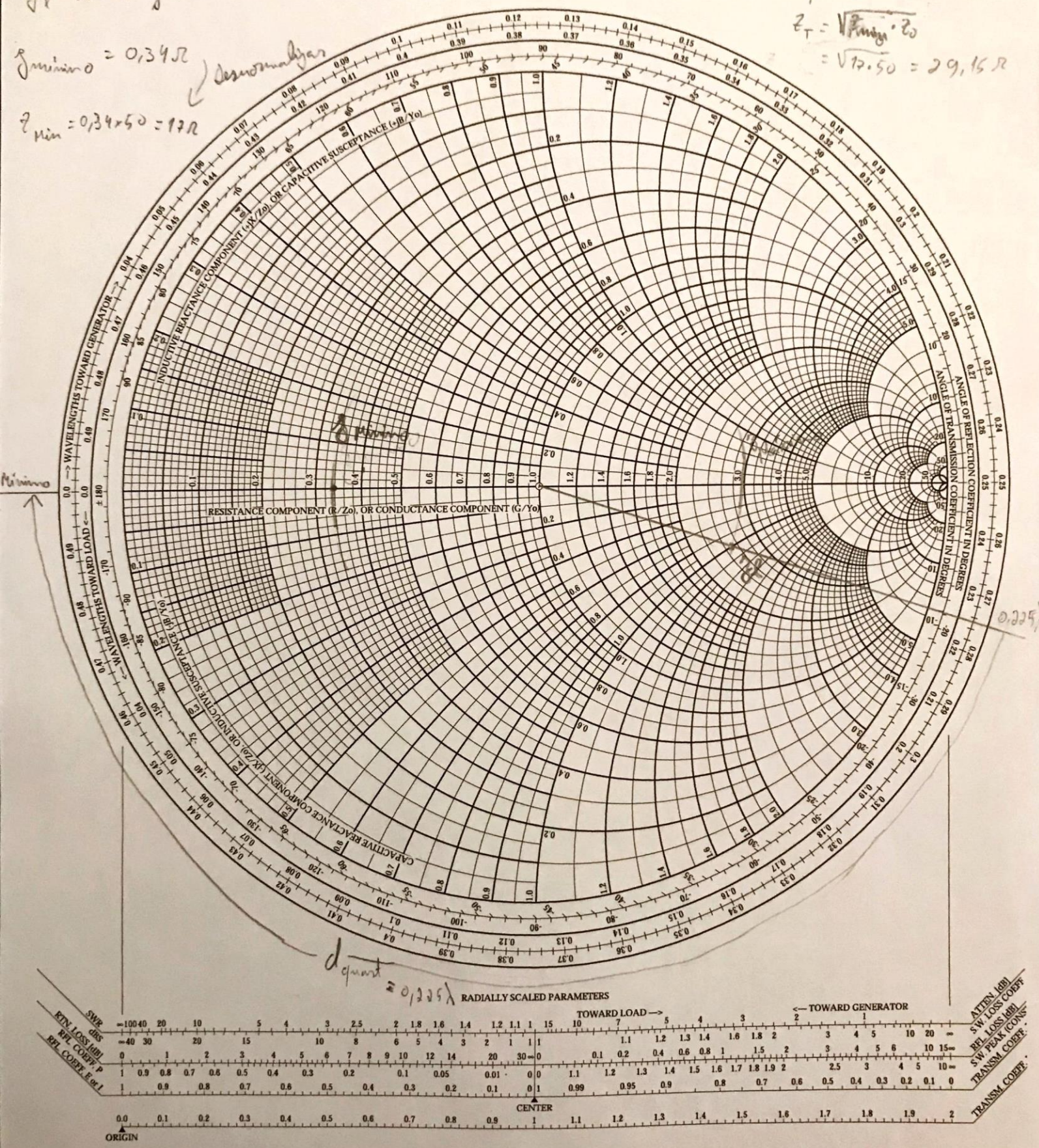


$Z_T = \sqrt{z_{min} \cdot Z_0}$

$= \sqrt{17 \cdot 50} = 29{,}15\,\Omega$

The Complete Smith Chart

Black Magic Design



Carga capacitiva → passamos por um mínimo e depois por um máximo